

Li Mingquan

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Education

Lanzhou University (985 / Top China University)

2023-present

B.S. in Microelectronics Science and Engineering

GPA:85.02/100

Awards

National Excellence Award, National Undergraduate Computer System Capability Competition (CPU Design)
Aug. 2026

National 3rd Prize, Fuwei Cup National College Student Electronic Design Competition Aug. 2026

Research interests

Computer Architecture & Microarchitecture, Digital VLSI Design

Project Experience

1. 32-bit Five-Stage Pipelined MIPS Processor

Sole Developer Mar.–Aug. 2026

Tech Stack: Verilog / Vivado / MIPS

- **Microarchitecture & Timing Optimization:** Designed an in-order 5-stage MIPS core supporting 47 instructions, precise exceptions, and a minimal SoC. Pruned critical path levels from 13 to 8 via FF-based Tag/Valid arrays, shared branch comparators ($\Delta t = -0.558$ ns), and fanout replication (net fanout: 7,714), boosting frequency from 75 to 117 MHz (+56%, IPC: 0.67).
- **Physical Bottleneck & Trace-Driven Pruning:** STA under 125 MHz revealed routing delay accounted for 69%–83% of critical paths (WNS = -0.342 ns), proving wire congestion was the real limit. Analyzed random memory access traces and intentionally discarded D-Cache (WNS = -1.086 ns) and SRAM prefetch (WNS = -1.134 ns) to avoid latency penalties with zero locality gain.
- **Silicon Debugging & Guardband Margin:** Resolved board-level failures absent in simulation (asynchronous SRAM read window limits and synthesis-induced cross-module MUX glitches). Backed off 10% from the post-CI frequency limit (WNS = $+0.107$ ns) to ensure a reliable ≥ 1.0 ns timing guardband on FPGA.

2. Intelligent Digital Security Sensor

Core Developer Jun.–Aug. 2026

Tech Stack: FPGA/Verilog / Vivado / APB Bus / CDC Clock Domain Crossing Design

- **Out-of-Band Hardware Loop & Sensing:** Built an autonomous defense circuit against clock glitches and voltage/thermal drift decoupled from CPU/bus arbitration. Combined a 51-stage RO sensor (20 ms window, < 0.5 ppm error vs. $\pm 0.5\%$ threshold) with a 16-stage TDL and a 5 ns pulse monitor (15--45 ns window) via OR-logic to eliminate blind spots.
- **Adversarial Hardening & Tamper Resistance:** Implemented a power-on baseline freeze strategy to thwart slow-drift ("boiling-frog") attacks. Integrated a 100 ms silent-failure watchdog and reset-cleared security latches (NMI/WIPE) to prevent compromised software from clearing tamper evidence.
- **FPGA Validation & Resource Closure:** Built automated testbenches for false-positive/negative rates and an FPGA testbed mixing XOR glitches via VIO. Closed timing across 4 clock domains at 200 MHz with minimal overhead (1,382 LUTs, 2,006 FFs, 0.223 W).

• Professional Skills

- Proficient in C, Python, and Verilog. Proficient in FPGA development and EDA tools.